**MAHATMA GANDHI UNIVERSITY, KOTTAYAM**

**MGU-UGP (HONOURS) Regular EXAMINATION October 2025**

**Third SEMESTER**

**Discipline Specific Elective (DSE) - MG3DSEFSQA00-INTRODUCTION TO ONBOARD FOOD SERVICE SYSTEMS**

(2024 ADMISSION ONWARDS)

Duration: 2 Hours Maximum Marks: 70

***Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(1) and Appreciation(Ap)***

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

**Part A**

MCQ

Answer all questions

Each question carries **1** marks

1. a) Freezing and dehydrating
2. a) Maintaining food quality at high altitude
3. c) Trains
4. a) By reducing hunger and stress
5. b) To enhance passenger experience during travel
6. a) Report and avoid using it
7. a) To prevent foodborne illness
8. a) To prepare and store meals safely
9. a) Check sanitation and hygiene
10. a) To reduce microbial contamination
11. b) To prevent foodborne illness
12. a) By providing easy access to food and drinks
13. a) Nutritional balance
14. c) Chapati and dal
15. a) Proper storage and temperature control
16. a) Due to limited sun exposure in space
17. a) Port Health inspection
18. b) Ensures microbial control and regulatory compliance
19. c) Limited by equipment and no open flame allowed
20. a) WHO sanitation code

**Part B**

Short Answer

Answer any **6** questions

Each question carries **5** marks

**21. Describe how food services onboard improve the overall travel experience**

- Provides comfort and satisfaction during long journeys.
- Enhances the airline's image and passenger loyalty.
- Offers refreshment and relaxation, reducing travel fatigue.
- Reflects hospitality and attention to customer care.
- Helps passengers manage meal timings across time zones.
- Creates a positive impression through taste, variety, and presentation.

**22. Describe the importance of personal hygiene of food handlers in flight catering units**

- Prevents contamination of food with harmful microorganisms.
- Ensures passenger safety and prevents foodborne illnesses.
- Maintains cleanliness standards required by aviation authorities.
- Promotes professionalism and builds customer trust.
- Includes practices such as clean uniforms, handwashing, hair covering, and health checks.
- Ensures compliance with HACCP and airline hygiene regulations.

**23. Explain how passenger feedback is incorporated into onboard catering services**

- They collect feedback through surveys, comment cards, and digital platforms.
- Data is analyzed to identify popular dishes and areas for improvement.
- Menus are revised based on taste preferences and dietary needs.
- Feedback helps in improving service timing, portion sizes, and packaging.
- Enhances overall meal quality and customer satisfaction.

**24. Demonstrate as to how the airlines accommodate diabetic passengers**

- Provide pre-booked special diabetic meals (low sugar, low fat, controlled carbohydrates).
- Use artificial sweeteners instead of sugar.
- Avoid fried or high-fat foods.
- Include whole grains, lean proteins, and fresh fruits/vegetables.
- Ensure meal timing matches the passenger's medical needs.
- Proper labeling and handling to prevent mix-ups.

**25. List two cultural benefits of offering regional dishes onboard**

- Promotes cultural diversity and inclusivity among passengers.
- Offers a sense of familiarity and comfort to travelers from different regions.
- Showcases the airline's respect for various cuisines and traditions.
- Enhances cultural exchange and appreciation during travel.

**26. Explain why food service is important for passengers on cruise ships**

- Long-duration journeys require proper nutrition and variety.
- Creates a major part of the entertainment and leisure experience.
- Reflects the cruise line's hospitality standards.
- Supports passenger health and comfort during the voyage.
- Special themed dinners and buffets add to enjoyment and satisfaction.

**27. Explain the function of cooking equipment in a pantry car**

- Used for reheating, cooking, and serving meals to passengers on trains.
- Includes ovens, boilers, fryers, and food warmers.
- Maintains food at safe serving temperatures.
- Ensures quick service during travel with limited space.
- Supports hygienic preparation and minimizes food spoilage.

**28. Differentiate between packaging methods used on Earth and in space**

| **Aspect** | **On Earth** | **In Space** |
|---|---|---|
| Atmosphere | Normal air pressure | Zero gravity and no atmosphere |
| Packaging Material | Plastic, paper, foil, trays | Lightweight, vacuum-sealed pouches |
| Purpose | Protection, convenience, presentation | Prevent spillage, easy handling in microgravity |
| Storage | Refrigerated or ambient | Compact, long shelf life, minimal waste |
| Heating Method | Microwave, stove, or oven | Specialized heaters (no open flame) |

## Part C

Essay

Answer any **2** questions

Each question carries **10** marks

**29. Analyze how hygiene and cleaning routines are planned on a cruise ship**

- Hygiene routines are scheduled daily and monitored by housekeeping and food safety teams.
- Cleaning plans follow international maritime health and sanitation standards.
- Different zones (kitchen, dining, cabins, decks) have specific cleaning schedules.
- Use of approved disinfectants and sanitizers for all food contact surfaces.
- Regular inspection of galleys, storage rooms, and waste disposal areas.
- Staff undergo hygiene training and follow handwashing and uniform protocols.

- Waste segregation and pest control systems ensure a clean environment.
- Records are maintained for audits and health authority inspections.

**30. Discuss the safety, hygiene, and sanitation measures incorporated in pantry car design**

- Pantry cars are designed with **stainless steel surfaces** for easy cleaning and corrosion resistance.
- **Adequate ventilation** and **temperature control** to prevent food spoilage.
- **Separate areas** for raw and cooked foods to avoid cross-contamination.
- Provision of **handwashing stations** and **waste bins** with covers.
- **Fire safety systems**, including extinguishers and alarms, installed for emergencies.
- Proper drainage and non-slip flooring to maintain sanitation.
- Regular cleaning schedules and pest control measures implemented.
- Compliance with **Indian Railways and FSSAI hygiene standards**.

**31. Discuss the definition and scope of onboard food service with suitable examples**

- **Definition:** Onboard food service refers to the preparation, handling, and serving of food and beverages to passengers while traveling by air, sea, or rail.
- **Scope:**
    - **Airlines:** In-flight catering units provide pre-packed meals like continental, vegetarian, or special diets.
    - **Cruise ships:** Offer elaborate multi-course meals, buffets, and specialty restaurants.
    - **Railways:** Pantry cars or station-based catering for passengers during travel.
    - Covers menu planning, storage, service methods, and passenger satisfaction.
    - Involves hygiene, safety, and logistics management in restricted environments.

**32. Explain how special meal planning enhances the overall food service system onboard**

- Meets the **dietary needs** of passengers (e.g., diabetic, vegetarian, gluten-free).
- Increases **passenger satisfaction** and comfort during travel.
- Demonstrates the airline's or cruise line's **attention to customer care**.
- Reduces risks of **allergic reactions** or medical emergencies.
- Reflects **cultural sensitivity** and inclusiveness.
- Helps build **brand image** and repeat customer loyalty.
- Encourages efficient meal management and accurate pre-order systems.